

Generalised Data Structures Library

Technology: C++ Programming

Project Overview:

This project is a **C++ library of generic data structures** that provides **object-oriented implementations** of both **linear and non-linear data structures**.

It offers **ready-to-use functionalities** for fundamental as well as advanced operations, designed in a **generic way** using templates so that they can be reused with **any data type**.

The library is designed to support **clean OOP principles**, modularity, and extensibility— making it suitable for both academic learning and real-world application development.

Key Features

- **Linear Data Structures**
 - Singly Linear Linked List
 - Singly Circular Linked List
 - Doubly Linear Linked List
 - Doubly Circular Linked List
 - Stack (LIFO)
 - Queue (FIFO)
- **Non-Linear Data Structures**
 - Binary Search Tree (BST) with insert, delete, traversal operations
- **Algorithms**
 - Searching (Linear Search, Binary Search etc)
 - Sorting (Bubble Sort, Selection Sort, Insertion Sort etc)
- **Generic Implementation**
 - Uses C++ templates for data type independence.
 - Same implementation works for integers, floats, strings, and custom objects.
- **Library Format**
 - Designed as a reusable C++ library that can be linked with client applications.

Learning Outcomes

- Mastery of **C++ Object-Oriented Programming (OOP) principles**.
- Strong foundation in **linear and non-linear data structures**.
- Implementation of **generic programming with templates**.
- Practical knowledge of **searching and sorting algorithms**.
- Experience in designing **reusable libraries** for software development.